

Thesis Appendix

JSPL EAF Industrial AI Decision Support — Release 1.0
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System summary

Industrial AI decision support for JSPL EAF tap-to-tap prediction and recipe optimization, integrating physics constraints, ML prediction (Phase 19), dual optimizers (20.2 + 31 V2), and hybrid trust framework (Phase 32).

Key results

Component	Result
Prediction test MAE	~2.1 min
Default 120 t prediction	~39.9 min
Hybrid reliability (live heats)	68-78 / 100
Optimizer V2	Planning-safe, no POWER optimization

Figures

Export from:

- ``research/phase_33_industrial_product_validation/thesis_figures/``
- ``research/phase_33_industrial_product_validation/publication_figures/``

Includes: SHAP importance, optimizer comparison, reliability distribution, digital twin roadmap.

Limitations

- Actual TTT not yet available for latest live heats
- Sensor gaps per Phase 27 limit digital twin readiness
- POWER recommendations from Phase 20.2 conflict with planning policy — use V2 for burden/flux only

Future work

See ``research/phase_33_industrial_product_validation/phase_34_future_work.md``